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Peer Review File

Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the *C. elegans* Box C/D complex

Corresponding Author: Mr Micha Birklbauer

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
In this manuscript, the authors developed a new software, MS Annika 3.0, for the identification of noncleavable cross-linked peptides. The authors argued that existing tools suffer from the large search space of the non-cleavable cross-linked peptides, and cannot search against proteome-wide databases. They developed MS Annika 3.0 to address this challenge. However, to the best of my knowledge, this problem has been solved years ago. There have been publications proposing algorithms to exhaustively search for all possible peptide-peptide pairs with linear time and space complexity (see the following major issue points). There are also tools, such as Kojak and pLink2, that do not perform exhaustive searches but have a fast speed. With the aforementioned publications and tools, the problem that this manuscript attempts to solve is no longer present. Furthermore, the algorithms proposed in this manuscript do not have significant innovation because fragment peak binning, sparse matrix multiplication, searching one chain and then pairing the other chain have been well studied by many publications in the past years.

Major issues
1. The authors failed to review or mention the existing publications, https://doi.org/10.1021/acs.jproteome.7b00338 and https://doi.org/10.1093/bioinformatics/bty526, that proposed the algorithms to exhaustively search for all possible peptide-pairs with linear computational complexity. The second publication also showed that they could search for 10000 proteins in a few minutes using a computer five years ago. While this manuscript showed that MS Annika 3.0 took hours with state-of-the-art computers.
2. Related to the previous point, the authors also failed to compare their software with the fastest ones Kojak (https://doi.org/10.1021/pr501321h) and Xolik (https://doi.org/10.1093/bioinformatics/bty526). In Section 2.4, the authors only compared the identifications, but not the runtime. Comparing the run time with other fast non-cleavable cross-linked peptides identification tools, including pLink2, Kojak, Xolik, and ECL2, is necessary because the major selling point of this manuscript is the fast search for proteome-wide databases.
3. The manuscript did not clearly describe whether MS Annika 3.0 performs an exhaustive search (i.e., search for all possible peptide-peptide pairs for each spectrum) or a non-exhaustive search (i.e., use heuristic methods to filter out certain peptides before searching and scoring them). The vague description seems to be a non-exhaustive search, which might result in missing findings.

Reviewer #2

(Remarks to the Author)
In the manuscript “Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the C. elegans Box C/D complex”, the authors introduce a novel software tool designed for global identification of non-cleavable crosslinked peptides. The analysis of non-cleavable crosslink data has been a challenge due to the extensive search space with all possible combinations of two peptides. The search algorithm enabled by sparse matrix calculations effectively reduced the computational time and provided good FDR compared to other existing software. This work can be significant in facilitating the studies on protein structures and interactions using crosslinking mass spectrometry. Meanwhile, the authors provided a clear explanation of the mathematical basis for this algorithm, and the effectiveness of the software is well-supported by experimental datasets from synthetic peptides. In this study, the authors used structural modeling via AlphaFold to examine the crosslink search results for C. elegans Box C/D Complex. However, we lack expertise in protein structural biology so were unable to provide comments for this part of the manuscript. The manuscript is well-organized and

provides sufficient information on the installation and use of MS Annika 3.0. Therefore, we recommend accepting this manuscript with minor revisions. Please consider the following points.

For the filtered Proteome, did you also keep decoy proteins with confident PSMs? Keeping only target proteins and eliminating decoy proteins would bias the results. Thank you for providing details for this search, search results, the complete c. elegans protein database and the filtered database in the repository.

The comparison against other search engines is comprehensive. Please provide a comparison of search times (in the supplement) as this was an important motivation for the MS Annika 3.0 algorithm

It makes sense that the requirement for diagnostic ions diminished the number of identified spectra. However, were rescoring algorithms such as percolator to come online in the future, report of the observation of diagnostic ions would be beneficial. Perhaps this could be mentioned somewhere.

Reviewer #3

(Remarks to the Author)
Birklbauer et al. introduce new functionalities into their software MS Annika, providing enhanced support for non-cleavable cross-links. The manuscript describes the underlying algorithm in detail and the software is evaluated with different data sets. These include a new data set from C. elegans nuclei cross-linked with DSG and some previously released benchmark data sets. Cross-links from the C. elegans experiment are used to improve AlphaFold2 predictions of the Box C/D RNP complex with the help of AlphaLink2.

Strengths:

Overall, this manuscript nicely describes new functionality in MS Annika that improves the analysis of data from experiments with non-cleavable cross-linking reagents such as DSG or ADH. The implementation of the algorithm and its benchmarking with synthetic data sets is discussed very clearly and the performance of the software seems to be at least on par with many competitors. Moreover, the C. elegans data set benefits from data reprocessing with the xiFDR tool from the Rappsilber lab, resulting in improved false discovery rate control.

Weaknesses:

The entire discussion of the C. clegans data, which is prominently featured in the title of the manuscript, could be improved. While the total number of cross-links is discussed repeatedly, there is no information about the number of inter-protein links or protein-protein interactions uncovered with this experiment. Moreover, no detailed results are given in the manuscript or its Supporting Information. I tried to look into some data deposited in PRIDE, but the repository was suffering from download issues, and it was not easily possible to find tables with cross-links. This is insofar important, as it does not become very clear why the Box C/D example was selected. Were there many other PPIs that could have been used for structure prediction?
The Box C/D example itself is not very strong. Four inter-protein links are used as the basis for structure prediction with AF2 or AL2. The increase in ipTM score for the cross-link-assisted model is notable, but small. Nevertheless, the entire architecture of the two models is so different that they cannot really be compared. The AF2 model fulfils one restraint, and a second one barely misses the imposed threshold. However, the distance of 29.5 Å for that second link is not excessively long given the many disordered regions in the complex. The AL2 model shows two compliant and two violated links, but one link that seemed to match in the AF2 model is now completely incompatible. A clear advantage for the AL2 model is not that obvious. In short, this system is maybe not the best showcase for the software, and the claim (at the end of the discussion section) that the model "significantly enhances our understanding of the assembly and functional dynamics of the complex" appears exaggerated. Consequently, I suggest focusing on the improvements in the software as the main message of the manuscript.

Other comments:

In the introduction, the authors discuss potential side reactions with more complex reagent designs. I am not aware that this is a huge problem. Same for long-term chemical stability, this would mostly affect the reactive sites such as the NHS ester group in lysine-reactive reagents, which are the same irrespective of the linker design. If the authors can cite any studies where the above points are discussed, they should be added. I think the bigger issue is that fewer reagents with cleavable linkers are commercially available. Moreover, some strategies such as reagents leading to zero-length links (EDC etc.) cannot be adapted to cleavable linkers.

Page 7: The authors mention "multiple" interaction sites between nol-56 and fib-1, which according to Fig. 3 are two sites on one protein and one site on the other. Multiple should be changed to two.

Although xiSEARCH showed promising performance with the benchmark data sets, why was this software not chosen to calculate overlaps in SI Figs. 4 and 5?

In general, the implementation of new functionalities in MS Annika are clearly described in Section 4.1, however, a few

points remain unclear:
-) MS Annika 3.0's non-cleavable search mode offers different matrix multiplication algorithms, as discussed in the text and in Table 1. The text mentions that this is a user-definable parameter, however it remains unclear what would be potentially different outcomes if different algorithms are selected, apart from different run times.
-) Page 15: What is meant by "peptidoform" here? Different localizations of the cross-linking site?
-) How is the beta peptide scored (or is it at all)?
-) In section 4.1, it is mentioned that by default, 10 alpha peptides are scored, while for the C. elegans study and in other cases, this value was apparently set to 2? Was this only done to reduce search times? Would this not increase the risk of missing candidates in larger search spaces?

Supplementary Tables 4 to 7 do not need to be provided in separate files, they can be integrated into the SI document.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have addressed most of my comments. From the authors' response, it seems that the speed, sparse matrix multiplication, or fragment peak binning are no longer innovations in this manuscript, but the FDR control is. Thus, the authors should shift the highlight to their FDR estimation method, which is described nowhere in the manuscript, and also revise the Figure 1 accordingly.

The following are my responses to some of the authors' responses.

Authors' response #2

Thank you for performing the detailed and thorough benchmarking. As Figure 6 shows, MS Annika 3.0 is not the fastest one among its competitors. Both pLink and Xolik have similar or higher speeds. However, MS Annika 3.0 has the best FDR estimation and control. Thus, the authors should highlight their FDR estimation method and describe in detail about the innovations/differences compared to the FDR methods of other tools.

Authors' response #3

As far as I know, the popular peptide identification tool, Comet, uses the sparse matrix multiplication. The method is published in https://doi.org/10.1007/s13361-015-1179-x. The authors should cite this paper. Regarding the engineering techniques, such as using the newer CPU instructions, to implement the method, the authors should also highlight and describe it in detail in the manuscript. It would be excellent if the authors could explain why their sparse matrix multiplication is better than the others if the authors think it is this manuscript's innovation and contribution.

I have no comments on the other responses.

Reviewer #2

(Remarks to the Author)
Excellent revision. Congratulations on a fine publication.

Reviewer #3

(Remarks to the Author)
In this revised version, the authors have addressed all my requests appropriately. I have no further comments.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have addressed all my comments.

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In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.
The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.
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Authors' Response Manuscript COMMSCHEM-24-0505-T "Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the C. elegans Box C/D complex"

We want to thank the reviewers for taking the time and effort to review our manuscript and provide detailed feedback.

Below follows a point-by-point response to the comments made by the reviewers. The original reviewer comments are followed by our response in blue. For convenience we have consecutively numbered reviewer statements and our answers.

Reviewers' comments:

Reviewer #1

(Remarks to the Author):

1: In this manuscript, the authors developed a new software, MS Annika 3.0, for the identification of noncleavable cross-linked peptides. The authors argued that existing tools suffer from the large search space of the non-cleavable cross-linked peptides, and cannot search against proteome-wide databases. They developed MS Annika 3.0 to address this challenge. However, to the best of my knowledge, this problem has been solved years ago. There have been publications proposing algorithms to exhaustively search for all possible peptide-peptide pairs with linear time and space complexity (see the following major issue points). There are also tools, such as Kojak and pLink2, that do not perform exhaustive searches but have a fast speed. With the aforementioned publications and tools, the problem that this manuscript attempts to solve is no longer present. Furthermore, the algorithms proposed in this manuscript do not have significant innovation because fragment peak binning, sparse matrix multiplication, searching one chain and then pairing the other chain have been well studied by many publications in the past years.

***1:** We thank the reviewer for their efforts and voicing their concerns about the novelty of our work as well as bringing to our attention that our original manuscript might benefit from additional information about the state of non-cleavable crosslink identification. Please find below our response to the key concerns raised:*

2: "However, to the best of my knowledge, this problem has been solved years ago. There have been publications proposing algorithms to exhaustively search for all possible peptide-peptide pairs with linear time and space complexity (see the following major issue points). There are also tools, such as Kojak and pLink2, that do not perform exhaustive searches but have a fast speed. With the aforementioned publications and tools, the problem that this manuscript attempts to solve is no longer present."

***2:** While we acknowledge that crosslink search engines which address this problem already exist, we disagree that this problem is "solved". As already stated in our original manuscript, pLink is limited by its inaccurate false discovery rate (FDR) estimation, reporting much more false positive crosslinks than estimated which hinders real world usage (see Figure 4 and 5). For a target FDR of 1% pLink yields an experimentally validated FDR of 6.52% for the benchmark dataset by Beveridge and co-workers [1] at an average of 242 true and 17 false positive crosslink identifications. On the contrary, our algorithm MS Annika only reports 3 false positives on average while still identifying 218 true crosslinks, therefore staying very close to the target FDR at 1.37% experimentally*

validated FDR. Similar is the case for the benchmark dataset by Matzinger and co-workers [2], where pLink reports on average 103 true and 15 false positive hits at an experimentally validated FDR of 11.85%, more than 10 percentage points off the target FDR. Again, MS Annika on the other hand only identifies 4 false positive crosslinks while still returning 89 true identifications on average, at a much better experimentally validated FDR of 3.92%. For convenience the numbers are also given in Table R1 and R2 (additionally to Figure 4 and 5 in the main manuscript). Kojak and Xolik are even weaker in that regard, heavily underestimating FDR (see also Figure 4 and 5). Kojak reports on average 48 false positive crosslinks for the dataset by Beveridge and co-workers [1], yielding an experimental FDR of 20.04%. This has already been reported by the original authors of the dataset [1] and we could confirm this also with the newest version of Kojak. For the dataset of Matzinger and co-workers [2] Kojak lacks in sensitivity as none of the identified crosslinks pass the 1% estimated FDR threshold and therefore reporting zero validated crosslinks. Xolik reports high numbers of false positive hits with experimentally validated FDRs ranging up to 71.03%. This especially highlights that usage of these tools has to be handled with extreme care in practice. Furthermore, Kojak is also considerably slower than both pLink and our algorithm (see Figure 6). Runtimes are also shown in condensed form in Table R3 for convenience. Most importantly, only a small number of publications exist that employ proteome-wide non-cleavable studies which also can be attributed to the associated computational challenges, showing that this problem has not yet been solved.

Tool	Crosslinks (true)	Crosslinks (false)	Experimental FDR
Kojak	189	48	20.04%
pLink	242	17	6.52%
Xolik	9	11	55.31%
MS Annika	218	3	1.37%

Table R1: *Comparison of Kojak, pLink, Xolik and MS Annika on the crosslink benchmark dataset by Beveridge and co-workers [1] as also used in our main manuscript. Crosslinks were validated for 1% estimated FDR. MS Annika is the only tool that yields accurate FDR estimates while reporting high numbers of true crosslinks. All numbers are averages from three technical replicates. This table is an excerpt from the results shown in Figure 4 of the main manuscript. Raw results are available via:*

https://github.com/hgb-bin-proteomics/MSAnnika_NC_Results/tree/master/Peplib_Beveridge

Tool	Crosslinks (true)	Crosslinks (false)	Experimental FDR
Kojak	0	0	Not Available
pLink	103	14	11.85%
Xolik	8	47	71.03%
MS Annika	89	4	3.92%

Table R2: *Comparison of Kojak, pLink, Xolik and MS Annika on the crosslink benchmark dataset by Matzinger and co-workers [2] as also used in our main manuscript. Crosslinks were again validated for 1% estimated FDR. Exactly as in the dataset by Beveridge and co-workers, MS Annika is the only of these four search engines that does not substantially underestimate FDR, only reporting 4 false positive crosslinks while pLink and Xolik report 14 and 47 respectively. For Kojak none of the identified crosslinks pass FDR validation and Kojak reports zero crosslinks at 1% estimated FDR. An experimentally validated FDR could therefore not be calculated for Kojak. All numbers are averages from three technical replicates. This table is an excerpt from the results shown in Figure 5 of the main manuscript. Raw results are available via:*

https://github.com/hgb-bin-proteomics/MSAnnika_NC_Results/tree/master/Peplib_Matzinger

Tool	Runtime (s)
Kojak	1229
pLink	258
Xolik	145
MS Annika	246

Table R3: *Comparison of runtimes of Kojak, pLink, Xolik and MS Annika using a human proteome-wide search. Xolik yields the shortest runtime of 145 seconds, while MS Annika takes 246 seconds, pLink 258 seconds and Kojak 1229 seconds on average. Numbers are averages from five consecutive runs. This table is an excerpt from the results shown in Figure 6 of the main manuscript. Raw results are available via:*

https://github.com/hgb-bin-proteomics/MSAnnika_NC_Results/tree/master/Runtimes

Actions taken: We have expanded the introduction, specifically mentioning other tools capable of proteome-wide non-cleavable crosslink identification and have added both Kojak and Xolik to all of our comparisons, significantly expanding our results (including Figure 4, Figure 5 and Figure 6). We also mention limitations and why we still consider this a relevant problem.

3: "Furthermore, the algorithms proposed in this manuscript do not have significant innovation because fragment peak binning, sparse matrix multiplication, searching one chain and then pairing the other chain have been well studied by many publications in the past years."

3: *As all research we are building on already known concepts and we specifically mention both fragment peak binning and the two-pass approach with suitable references in our manuscript, giving credit to scientists who have explored these topics before. However, to the best of our knowledge sparse matrix multiplication using a matrix encoded peptide database and matrix encoded spectra is not something that has been done before, neither in crosslinking nor proteomics in general. We would be happy to reference related work if the reviewer can provide a fitting publication. Furthermore, in the rare case that there is other related work we are convinced that the differences are going to be significant as matrix optimization as a research field has moved fast in the past years and hardware-based support for newer CPU instruction sets has only reached broad adoption very recently, e.g. in 2022 for AVX-512 with the release of Zen 4 CPUs by AMD (see https://www.amd.com/content/dam/amd/en/documents/epyc-business-docs/white-papers/221704010-B_en_4th-Gen-AMD-EPYC-Processor-Architecture---White-Paper_pdf.pdf, last accessed 09.11.2024). Our implementation is based on Eigen and Nvidia CUDA cuSPARSE which leverage newest hardware and software optimizations for matrix multiplication if available. Lastly, since our algorithm is also based on our own in-house developed and previously published peptide search engine MS Amanda [3,4,5] the results would always be different to any other implementation using a similar method.*

Major issues

4: The authors failed to review or mention the existing publications, https://doi.org/10.1021/acs.jproteome.7b00338 and https://doi.org/10.1093/bioinformatics/bty526, that proposed the algorithms to exhaustively search for all possible peptide-pairs with linear computational complexity. The second publication also showed that they could search for 10000 proteins in a few minutes using a computer five years ago. While this manuscript showed that MS Annika 3.0 took hours with state-of-the-art computers.

4: *We thank the reviewer for bringing these publications to our attention. We have included a paragraph mentioning this tool (Xolik) in the introduction. Moreover, even though a direct comparison between the times reported by the authors of Xolik and our manuscript is not possible as the problems are considerably different, we recognize that an in-depth analysis of runtimes is something that we missed in our original draft. We have added this to our manuscript and show that MS Annika outperforms Kojak and pLink in terms of search speed, but is significantly slower than Xolik (see Figure 6, and answer 2). While Xolik shows an impressive performance with regard to search speed, MS Annika can solve different problems: It supports other (custom) digestion enzymes aside of trypsin, as well as multiple possible crosslink sites or n-terminal modifications. We are therefore convinced that MS Annika will be highly useful and a complementary contribution to the landscape of existing crosslink tool such as Xolik.*

Actions taken: We have added a paragraph in the introduction specifically mentioning Xolik and expanded the results by adding Xolik to all our comparisons (Figure 4 and Figure 5). Furthermore, we have conducted an in-depth analysis of crosslink search engine runtimes comparing our algorithm to Kojak, pLink and Xolik (Figure 6, Supplementary Figure 8 and Supplementary Figure 9).

5: Related to the previous point, the authors also failed to compare their software with the fastest ones Kojak (https://doi.org/10.1021/pr501321h) and Xolik (https://doi.org/10.1093/bioinformatics/bty526). In Section 2.4, the authors only compared the identifications, but not the runtime. Comparing the run time with other fast non-cleavable cross-linked peptides identification tools, including pLink2, Kojak, Xolik, and ECL2, is necessary because the major selling point of this manuscript is the fast search for proteome-wide databases.

5: *We thank the reviewer for giving input on what they would prefer to see in comparisons. With over 50+ crosslink search engines it is unfortunately unfeasible to compare against every single one. We recognize the importance of comparing both analytical as well as computational performance for Kojak and Xolik. We did not include ECL2 because – as far as we understood it – Xolik incorporates an updated version of the ECL2 algorithm. As mentioned in the answers to the previous points we have included Kojak and Xolik in all our comparisons, assessing sensitivity, robustness of FDR estimation and computational performance (see Figure 4 – 6). We show that our algorithm outperforms both Kojak and Xolik in terms of number of identifications as well as FDR estimation, reporting a lot less false positives than Kojak and Xolik (see Figure 4 and 5, and answer 2). We also show that MS Annika has shorter search times than Kojak (see Figure 6).*

Actions taken: We have significantly expanded the results, adding Kojak and Xolik to all comparisons (Figure 4 and Figure 5). We also included separate chapter on computational performance with an expansive investigation of search engine runtimes (Figure 6, Supplementary Figure 8 and Supplementary Figure 9).

6: The manuscript did not clearly describe whether MS Annika 3.0 performs an exhaustive search (i.e., search for all possible peptide-peptide pairs for each spectrum) or a non-exhaustive search (i.e., use heuristic methods to filter out certain peptides before searching and scoring them). The vague description seems to be a non-exhaustive search, which might result in missing findings.

6: *We apologize that we did not explicitly state this in our manuscript previously and have added the information that our algorithm uses a non-exhaustive approach at various suitable locations throughout our manuscript. We agree that non-exhaustive approaches always risk missing true identifications and we have set out to explore this further in the current version of our manuscript.*

We did not mention this in the original version of the manuscript, but MS Annika also offers to run exhaustive searches via a user-definable parameter. We compared the exhaustive search with the non-exhaustive search and found that the non-exhaustive approach outperforms the exhaustive one when it comes to validated results. The exhaustive approach identifies on average 215 true positive crosslinks at 1% estimated FDR for the dataset by Beveridge and co-workers [1] while the non-exhaustive approach reports 218 true positive crosslinks on average. Both approaches report 3 false positives, staying close to the 1% target FDR (see Supplementary Figure 7). We argue that this is due to the fact that an exhaustive approach considers all possible peptide-pairs for search which increases the chances of matching a random false positive hit or decoy match. This has to be accounted for during validation, leading to an increase of score cut-off to remain at the designated target FDR and in turn a loss in identifications that pass the target FDR threshold. Contrary the non-exhaustive approach only considers likely peptide combinations and therefore does not suffer from this limitation. Moreover, we show that our non-exhaustive algorithm outperforms other exhaustive approaches such as XlinkX and Xolik.

Actions taken: We have added the information of doing a non-exhaustive approach at various locations throughout the manuscript. We have expanded the result section (specifically section 2.4) with a detailed comparison of exhaustive and non-exhaustive search with additional supplementary material (Supplementary Figure 7 and Supplementary Table 3).

References:

[1] Beveridge, R., Stadlmann, J., Penninger, J. M. & Mechtler, K. A synthetic peptide library for benchmarking crosslinking-mass spectrometry search engines for proteins and protein complexes. Nature Communications 11 (2020). URL http://dx.doi.org/10.1038/s41467-020-14608-2.

[2] Matzinger, M. et al. Mimicked synthetic ribosomal protein complex for bench-marking crosslinking mass spectrometry workflows. Nature Communications 13 (2022). URL http://dx.doi.org/10.1038/s41467-022-31701-w

[3] Dorfer, V. et al. Ms amanda, a universal identification algorithm optimized for high accuracy tandem mass spectra. Journal of Proteome Research 13, 3679–3684 (2014). URL http://dx.doi.org/10.1021/pr500202e.

[4] Dorfer, V., Strobl, M., Winkler, S. & Mechtler, K. Ms amanda 2.0: Advancements in the standalone implementation. Rapid Communications in Mass Spectrometry 35 (2021). URL http://dx.doi.org/10.1002/rcm.9088.

[5] Buur, L. M. et al. Ms2rescore 3.0 is a modular, flexible, and user-friendly platform to boost peptide identifications, as showcased with ms amanda 3.0. Journal of Proteome Research 23, 3200–3207 (2024). URL http://dx.doi.org/10.1021/acs.jproteome.3c00785.

Reviewer #2

(Remarks to the Author):

1: In the manuscript "Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the C. elegans Box C/D complex", the authors introduce a novel software tool designed for global identification of non-cleavable crosslinked peptides. The analysis of non-cleavable crosslink data has been a challenge due to the extensive search space with all possible combinations of two peptides. The search algorithm enabled by sparse matrix calculations effectively reduced the computational time and provided good FDR compared to other existing software. This work can be significant in facilitating the studies on protein structures and interactions using crosslinking mass spectrometry. Meanwhile, the authors provided a clear explanation of the mathematical basis for this algorithm, and the effectiveness of the software is well-supported by experimental datasets from synthetic peptides. In this study, the authors used structural modeling via AlphaFold to examine the crosslink search results for C. elegans Box C/D Complex. However, we lack expertise in protein structural biology so were unable to provide comments for this part of the manuscript. The manuscript is well-organized and provides sufficient information on the installation and use of MS Annika 3.0. Therefore, we recommend accepting this manuscript with minor revisions. Please consider the following points.

***1:** We thank the reviewer for taking the time to assess our submission and the overall positive feedback!*

2: For the filtered Proteome, did you also keep decoy proteins with confident PSMs? Keeping only target proteins and eliminating decoy proteins would bias the results. Thank you for providing details for this search, search results, the complete c. elegans protein database and the filtered database in the repository.

***2:** We thank the reviewer for bringing this to our attention. We originally did not consider decoy PSMs for the creation of the filtered database and would have missed this. We have re-constructed the filtered database, this time including decoy PSMs and have re-run all affected crosslink searches. The inclusion of decoy PSMs resulted in an additional ~200 proteins being added to the filtered database going from 3069 to 3266 proteins. Crosslink results marginally improved across the three biological replicates: +36 crosslinks for replicate one, +9 crosslinks for replicate two, and -5 crosslinks for replicate three (all at 1% estimated FDR).*

Actions taken: We have re-constructed the filtered protein database including decoys. We have added this information to the methods section. Moreover, we have re-run all affected searches and updated the results, specifically Figure 2 and Supplementary Figures 1-3.

3: The comparison against other search engines is comprehensive. Please provide a comparison of search times (in the supplement) as this was an important motivation for the MS Annika 3.0 algorithm

***3:** Since this was requested by several reviewers we have included the comparison of runtimes in the main text. In short, we could show that MS Annika takes on average 4 minutes and 6 seconds for a human proteome-wide search of ~5200 mass spectra, outperforming both pLink and Kojak in search time. MS Annika falls short to Xolik in terms of search time, however due to the poor analytical performance of Xolik (as shown in Figure 4 and Figure 5) we still consider MS Annika the superior choice, unifying both analytical and computational performance.*

Actions taken: We added section 2.5 and Figure 6 to the results, giving an extensive overview of the runtimes of the different tools. We also added Supplementary Figure 8 and Supplementary Figure 9 with additional information on runtimes.

4: It makes sense that the requirement for diagnostic ions diminished the number of identified spectra. However, were rescoring algorithms such as percolator to come online in the future, report of the observation of diagnostic ions would be beneficial. Perhaps this could be mentioned somewhere.

4: *We appreciate this suggestion and agree.*

Actions taken: We have added a small paragraph about inclusion of diagnostic ions as potential features for rescoring to the discussion.

Reviewer #3

(Remarks to the Author):

1: Birklbauer et al. introduce new functionalities into their software MS Annika, providing enhanced support for non-cleavable cross-links. The manuscript describes the underlying algorithm in detail and the software is evaluated with different data sets. These include a new data set from C. elegans nuclei cross-linked with DSG and some previously released benchmark data sets. Cross-links from the C. elegans experiment are used to improve AlphaFold2 predictions of the Box C/D RNP complex with the help of AlphaLink2.

Strengths:

Overall, this manuscript nicely describes new functionality in MS Annika that improves the analysis of data from experiments with non-cleavable cross-linking reagents such as DSG or ADH. The implementation of the algorithm and its benchmarking with synthetic data sets is discussed very clearly and the performance of the software seems to be at least on par with many competitors. Moreover, the C. elegans data set benefits from data reprocessing with the xiFDR tool from the Rappsilber lab, resulting in improved false discovery rate control.

1: *We very much appreciate this positive assessment of our work and thank the reviewer for their efforts.*

Weaknesses:

2: The entire discussion of the C. clegans data, which is prominently featured in the title of the manuscript, could be improved. While the total number of cross-links is discussed repeatedly, there is no information about the number of inter-protein links or protein-protein interactions uncovered with this experiment. Moreover, no detailed results are given in the manuscript or its Supporting Information. I tried to look into some data deposited in PRIDE, but the repository was suffering from download issues, and it was not easily possible to find tables with cross-links. This is insofar important, as it does not become very clear why the Box C/D example was selected. Were there many other PPIs that could have been used for structure prediction?

2: *We apologize that we failed to include this indeed very important information in the initial submission of our manuscript. We have added two additional tables in the main manuscript outlining the number of inter and intra crosslinks across the biological replicates as well as intra and inter protein-protein interactions. In total we found 354 intra crosslinks and 81 inter crosslinks at 1% residue pair FDR and 192 intra protein PPIs as well as 52 inter protein PPIs at 1% PPI FDR. We added additional paragraphs discussing these numbers and also included a PPI network visualizing inter crosslinks that satisfy 1% PPI FDR. Moreover, we highlighted proteins of the Box C/D RNP complex in the network. This complex was selected due to several reasons: 1) it is well described in literature making comparison of our findings with established knowledge easier, 2) a large portion of PPIs in our network is ambiguous as crosslinks consist of peptides that are shared among several proteins, and 3) we have been extensively working with C. elegans and the complex was of interest to our research group. We are sorry that there were download issues with PRIDE, however that is unfortunately out of our control, but we can provide the identified crosslinks as a supplementary file!*

Actions taken: We have extended the corresponding result section adding more detailed information about the identified crosslinks and added Table 1 and Table 2 describing inter

and inter crosslinks and PPIs. We added a PPI network highlighting the Box C/D RNP complex as Supplementary Figure 4. Furthermore, we have provided the identified crosslinks as a supplementary file.

3: The Box C/D example itself is not very strong. Four inter-protein links are used as the basis for structure prediction with AF2 or AL2. The increase in ipTM score for the cross-link-assisted model is notable, but small. Nevertheless, the entire architecture of the two models is so different that they cannot really be compared. The AF2 model fulfils one restraint, and a second one barely misses the imposed threshold. However, the distance of 29.5 Å for that second link is not excessively long given the many disordered regions in the complex. The AL2 model shows two compliant and two violated links, but one link that seemed to match in the AF2 model is now completely incompatible. A clear advantage for the AL2 model is not that obvious.

3: *We argue that the two crosslinks that violate the distance constraints in the AL2 model are due to the inherent symmetry of the complex that unfortunately cannot be accounted for due to the limited number of complex members that are supported by AF2/AL2 for structural modelling. The long-distance crosslink between M28.5 and nol-56 could be attributed to a second M28.5 protein interacting with the C-terminal region of nol-56, similar to the predicted interaction between M28.5 and nol-58, which would satisfy the crosslink distance limit. Moreover, the same symmetrical consideration would apply to the long-distance link between fib-1and nol-58. Nevertheless, we recognize the concern of how much this is an improvement over the AF2 structure and consequently have rephrased the results to be more neutral.*

Actions taken: Usage of more neutral language throughout the structural analysis section.

4: In short, this system is maybe not the best showcase for the software, and the claim (at the end of the discussion section) that the model "significantly enhances our understanding of the assembly and functional dynamics of the complex" appears exaggerated. Consequently, I suggest focusing on the improvements in the software as the main message of the manuscript.

4: *We thank the reviewer for this suggestion. We have adopted a more neutral language throughout the manuscript and have changed abstract and discussion to set more realistic expectations about the described details of the Box C/D RNP complex. We have also further extended the manuscript with additional information on the software, as also requested by other reviewers.*

Actions taken: Adoption of abstract, results and discussion as well as the use of more neutral language in conjunction with the structural analysis. Several additions to the main text about the improvements of the software that should further shift the focus (comparison to more crosslink search engines, comparison of exhaustive and non-exhaustive search, comparison of runtimes).

Other comments:

5: In the introduction, the authors discuss potential side reactions with more complex reagent designs. I am not aware that this is a huge problem. Same for long-term chemical stability, this would mostly affect the reactive sites such as the NHS ester group in lysine-reactive reagents, which are the same irrespective of the linker design. If the authors can cite any studies where the above points are discussed, they should be added. I think the bigger issue is that fewer reagents with cleavable linkers are commercially available. Moreover, some strategies such as reagents leading to zero-length links (EDC etc.) cannot be adapted to cleavable linkers.

5: *We apologize for not providing a reference for said points. These are based on our own experience from one of our latest projects for which we currently have a manuscript in submission. We have a pre-print now available and referenced that accordingly.*

Actions taken: We have added the appropriate references to the manuscript.

6: Page 7: The authors mention "multiple" interaction sites between nol-56 and fib-1, which according to Fig. 3 are two sites on one protein and one site on the other. Multiple should be changed to two.

Actions taken: We have changed the "multiple" to "two".

7: Although xiSEARCH showed promising performance with the benchmark data sets, why was this software not chosen to calculate overlaps in SI Figs. 4 and 5?

7: *The answer is already in the question: in our experience xiSEARCH (in conjunction with xiFDR) generally provides accurate and robust results, as also reflected in the results on benchmark datasets. Moreover, especially for the dataset by Beveridge and co-workers [1] xiSEARCH reports similar numbers to MS Annika. We therefore thought it would be more interesting to look at overlaps between search engines that behave differently (or at least show different numbers of crosslink identifications). We did not include xiSEARCH as a fourth because Venn diagrams of more than three sets cannot be accurately scaled anymore. However, please find the overlaps of MS Annika and xiSEARCH/xiFDR below/on the next page:*

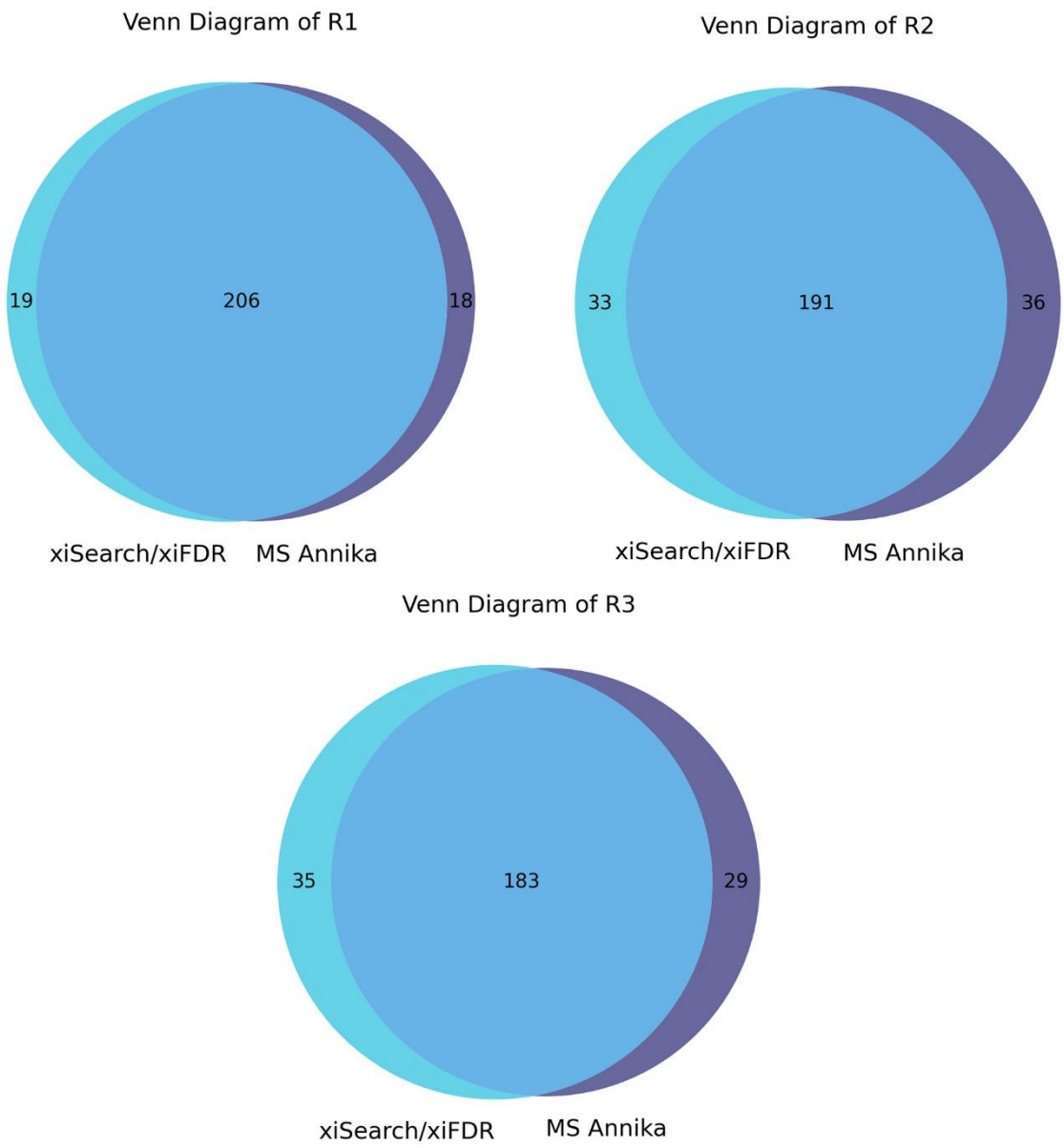


Figure R1: *Overlaps of crosslink identifications for xiSEARCH and MS Annika using the dataset by Beveridge and co-workers at 1% estimated FDR. Top left depicts the Venn diagram for technical replicate one, top right for replicate two and bottom middle for replicate three. The overlaps are in-line with what we also observed for other crosslink search engines (see Supplementary Figure 5).*

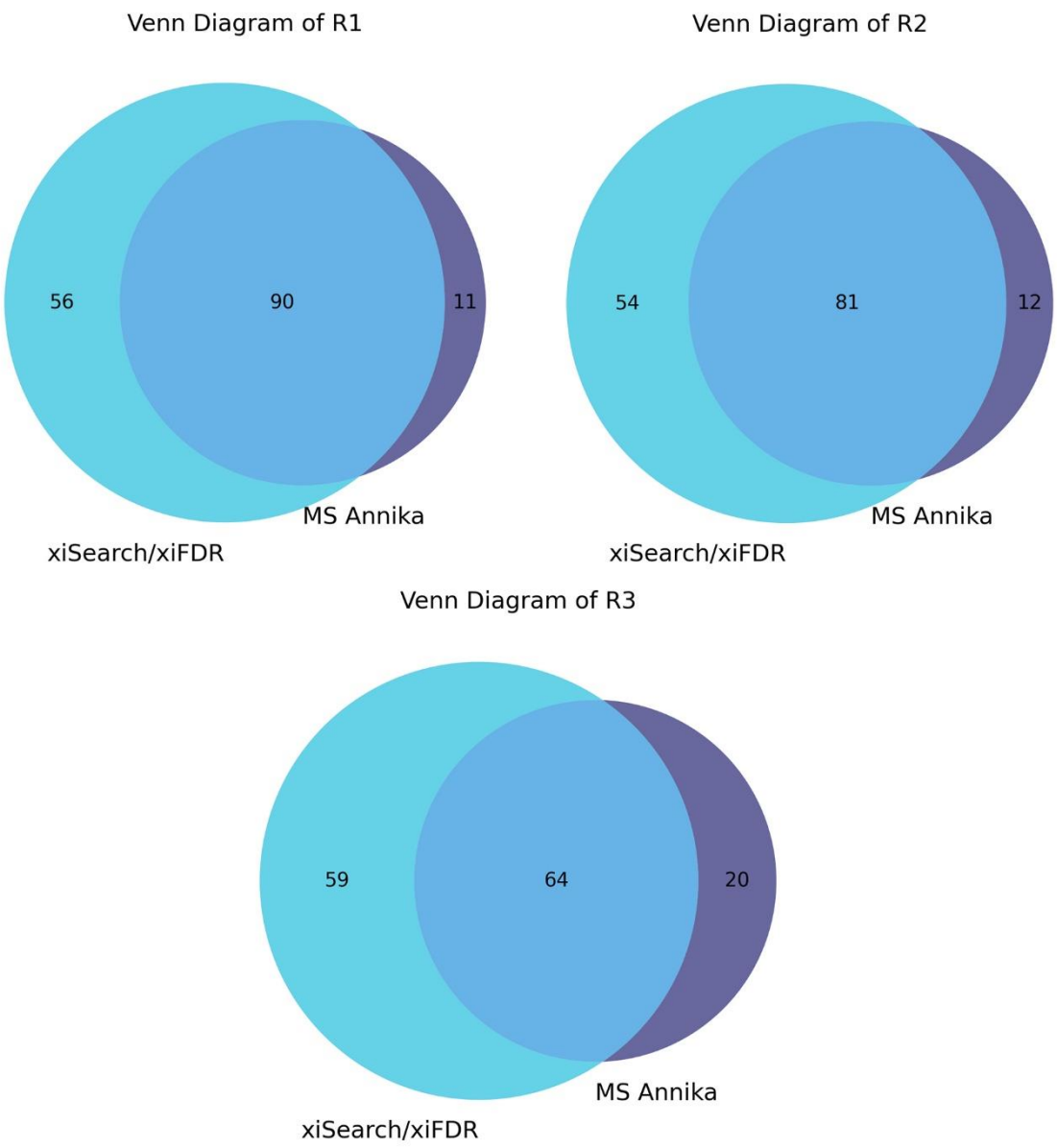


***Figure R2**: Overlaps of crosslink identifications for xiSEARCH and MS Annika using the dataset by Matzinger and co-workers at 1% estimated FDR. Top left depicts the Venn diagram for technical replicate one, top right for replicate two and bottom middle for replicate three. Here it's clearly visible that xiSEARCH outperforms MS Annika in terms of total crosslink identifications. The overlaps are again in-line with what we also observed for other crosslink search engines (see Supplementary Figure 6).*

8: In general, the implementation of new functionalities in MS Annika are clearly described in Section 4.1, however, a few points remain unclear:

-) MS Annika 3.0's non-cleavable search mode offers different matrix multiplication algorithms, as discussed in the text and in Table 1. The text mentions that this is a user-definable parameter, however it remains unclear what would be potentially different outcomes if different algorithms are selected, apart from different run times.

***8:** We thank the reviewer for bringing this to our attention. We failed to mention that all these algorithms yield identical results and only differ in performance which is dependent on hardware. We generally recommend the default parameter selection, however as shown in Figure 6*

switching to a GPU-based approach can be beneficial (if supported hardware is available), especially when searching a large number of mass spectra.

Actions taken: We added the information that all algorithms yield identical results to the manuscript.

9: -) Page 15: What is meant by "peptidoform" here? Different localizations of the cross-linking site?

9: *Exactly, different localizations of the crosslinking site as well as other potential post-translational-modifications.*

Actions taken: We added a specification for peptidoform at the appropriate place in the manuscript.

10: -) How is the beta peptide scored (or is it at all)?

10: *The beta peptide is scored using MS Amanda [2,3,4], our in-house developed peptide search engine that we also use to calculate an accurate score for the alpha peptide. Thanks for pointing out that we forgot to add that information!*

Action taken: We added information about scoring of the beta peptide to the method section.

11: -) In section 4.1, it is mentioned that by default, 10 alpha peptides are scored, while for the C. elegans study and in other cases, this value was apparently set to 2? Was this only done to reduce search times? Would this not increase the risk of missing candidates in larger search spaces?

11: *Over the course of testing MS Annika we have tried many different values for this parameter both on the dataset by Lenz and co-workers as well as our own C. elegans dataset and came to the conclusion that smaller values favor better results for large scale studies. The risk of missing true identifications increases, however at the same time the chance of matching a false positive also decreases (due to not considering lower scoring alpha peptides) which in our experience ultimately led to better results both in terms of number of identifications but also experimentally validated FDR. This is in many ways similar to the newly added discussion in section 2.4 of our manuscript that compares the non-exhaustive and exhaustive search. The reduction of search time is an added benefit.*

12: Supplementary Tables 4 to 7 do not need to be provided in separate files, they can be integrated into the SI document.

12: *We originally intended to submit Supplementary Tables 4 – 7 (now Supplementary Files 2-5) as "real" tables in either .xlsx or .csv format but they were automatically converted to PDF by the manuscript submission system. The tables are too large to be properly displayed in the Springer Nature LaTeX Template which we use for both the manuscript and the supplementary. If it is the wish of reviewers and editors we will look into integrating it directly into the supplementary, however we think that – especially now that we compare against two additional search engines – the tables are easier to navigate as separate files.*

References:

[1] Beveridge, R., Stadlmann, J., Penninger, J. M. & Mechtler, K. A synthetic peptide library for benchmarking crosslinking-mass spectrometry search engines for proteins and protein complexes. Nature Communications 11 (2020). URL http://dx.doi.org/10.1038/s41467-020-14608-2.

[2] Dorfer, V. et al. Ms amanda, a universal identification algorithm optimized for high accuracy tandem mass spectra. Journal of Proteome Research 13, 3679–3684 (2014). URL http://dx.doi.org/10.1021/pr500202e.

[3] Dorfer, V., Strobl, M., Winkler, S. & Mechtler, K. Ms amanda 2.0: Advancements in the standalone implementation. Rapid Communications in Mass Spectrometry 35 (2021). URL http://dx.doi.org/10.1002/rcm.9088.

[4] Buur, L. M. et al. Ms2rescore 3.0 is a modular, flexible, and user-friendly platform to boost peptide identifications, as showcased with ms amanda 3.0. Journal of Proteome Research 23, 3200–3207 (2024). URL http://dx.doi.org/10.1021/acs.jproteome.3c00785.

Authors' Response Manuscript COMMSCHEM-24-0505A-Z "Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the C. elegans Box C/D complex"

We thank the reviewers once more for taking the time and effort to review the revised version of our manuscript and provide detailed feedback.

Below follows a point-by-point response to the comments made by the reviewers. The original reviewer comments are followed by our response in blue. For convenience we have consecutively numbered reviewer statements and our answers.

Reviewers' comments:

Reviewer #1

(Remarks to the Author):

1: The authors have addressed most of my comments. From the authors' response, it seems that the speed, sparse matrix multiplication, or fragment peak binning are no longer innovations in this manuscript, but the FDR control is. Thus, the authors should shift the highlight to their FDR estimation method, which is described nowhere in the manuscript, and also revise the Figure 1 accordingly.

1: *We once more thank the reviewer for reading our revised manuscript and giving detailed feedback on the implemented changes. We thank the reviewer for suggesting to put additional focus on the FDR estimation which is indeed one of the strong parts of MS Annika. We have previously published our validation workflow [1] but recognize that this is still an integral part to the non-cleavable crosslink identification workflow and have therefore added an additional paragraph to the methods section that gives a summary of FDR estimation in MS Annika. Moreover, we have also included this step in Figure 1. However, we would still like to emphasize that we also still think that our approach to sparse matrix multiplication is innovative especially in the context of crosslinking, please consider our response to point 3 which elaborates more on this part.*

Actions taken: We have expanded the methods section giving a detailed description of how FDR is estimated in MS Annika and adapted Figure 1 to include this step as part of the workflow.

The following are my responses to some of the authors' responses.

2: Authors' response #2

Thank you for performing the detailed and thorough benchmarking. As Figure 6 shows, MS Annika 3.0 is not the fastest one among its competitors. Both pLink and Xolik have similar or higher speeds. However, MS Annika 3.0 has the best FDR estimation and control. Thus, the authors should highlight their FDR estimation method and describe in detail about the innovations/differences compared to the FDR methods of other tools.

2: We thank the reviewer for this assessment of our work. As mentioned in the previous point (1) we have extended the methods section, including a description of the FDR estimation algorithm in MS Annika. It should be noted that the approach that we employ for FDR estimation is a traditional target-decoy based method which is also used across other crosslink search engines, therefore the FDR algorithm in itself is most likely not the cause of the robust performance of

MS Annika. Rather the combination of the MS Annika CSM and crosslink score which are highly reflective of match quality and allow for good differentiation between true and false identifications, together with this FDR estimation algorithm, is what yields this consistent performance. The MS Annika CSM and crosslink scores are described in the methods section of this manuscript and are derivatives of our in-house developed MS Amanda score which is described in [2] (also referenced in the manuscript). Direct comparison between the FDR estimation in MS Annika and other tools is challenging as there is limited information available on how exactly other search engines handle validation beyond using a target-decoy based approach. We argue that the same concept holds true for other search engines as well, namely that there is strong interplay between the search engine score and the FDR estimation algorithm. Beyond that, one possible reason for the differences in FDR estimation between the various tools is that they validate results at different stages: Fischer et al. [3] reported that results need to be validated at each aggregation level for robust FDR estimation which most of these tools do not account for. Contrary, MS Annika validates results at CSM and crosslink level yielding this robust FDR estimation performance. Moreover, MS Annika also supports external validation with tools like xiFDR [3] which allow for even better FDR control as we also show in the manuscript.

Actions taken: We have expanded the methods section elaborating on the FDR estimation in MS Annika. We also adapted Figure 1 to include this crucial step in the overview of the MS Annika workflow.

3: Authors' response #3

As far as I know, the popular peptide identification tool, Comet, uses the sparse matrix multiplication. The method is published in https://doi.org/10.1007/s13361-015-1179-x. The authors should cite this paper. Regarding the engineering techniques, such as using the newer CPU instructions, to implement the method, the authors should also highlight and describe it in detail in the manuscript. It would be excellent if the authors could explain why their sparse matrix multiplication is better than the others if the authors think it is this manuscript's innovation and contribution.

I have no comments on the other responses.

***3:** We thank the reviewer for their input. We already mentioned (and cited) Comet in our original manuscript, specifically in the methods section where we describe our algorithm in detail which – to us – was the most appropriate place for this information. It is true that Comet uses sparse matrices for calculation of the cross-correlation score, however that is significantly different to what we do. We would like to emphasize the following differences:*

- *Comet encodes one mass spectrum as one sparse matrix*
- *MS Annika encodes one mass spectrum as one sparse vector and therefore multiple mass spectra as one sparse matrix*
- *Comet uses the sparse matrix as a fast and memory efficient data structure for accessing intensity values of the mass spectrum's peaks*
- *MS Annika does not store intensity values in the sparse matrix and uses the matrix for its mathematical properties, specifically for fast matrix multiplication*
- *Comet does not do matrix multiplication, the matrix is used to efficiently retrieve intensity values for specific fragment ion m/z values to calculate the cross-correlation score*
- *MS Annika does employ matrix multiplication for calculation of the score that we describe in the manuscript*

We hope that this highlights that even though both tools use sparse matrices, their approaches have very little in common. Moreover, for leveraging new CPU instructions sets and state-of-the-art hardware, we employ the C++ libraries Eigen (https://eigen.tuxfamily.org/) and Nvidia CUDA cuSPARSE (https://docs.nvidia.com/cuda/cusparse/) which we also state in the implementation section in our manuscript's methods. We did not implement sparse matrix multiplication on a low level ourselves as none of us is working in computational linear algebra, we apologize if it came across like that in our last response! We have referenced both of these tools accordingly in our manuscript as well as giving acknowledgement to the maintainers and community of these libraries.

References

[1] Pirklbauer, G. J. et al. Ms annika: A new cross-linking search engine. Journal of Proteome Research 20, 2560–2569 (2021). URL http://dx.doi.org/10.1021/acs.jproteome.0c01000.

[2] Dorfer, V. et al. Ms amanda, a universal identification algorithm optimized for high accuracy tandem mass spectra. Journal of Proteome Research 13, 3679–3684 (2014). URL http://dx.doi.org/10.1021/pr500202e.

[3] Fischer, L. & Rappsilber, J. Quirks of error estimation in cross-linking/mass spectrometry. Analytical Chemistry 89, 3829–3833 (2017). URL http://dx.doi.org/10.1021/acs.analchem.6b03745.

Reviewer #2

(Remarks to the Author):

Excellent revision. Congratulations on a fine publication.

We thank the reviewer for their time and effort as well as the positive assessment of our work!

Reviewer #3

(Remarks to the Author):

In this revised version, the authors have addressed all my requests appropriately. I have no further comments.

We thank the reviewer for taking the time and effort to evaluate our revised manuscript and their positive assessment!

Authors' Response Manuscript COMMSCHEM-24-0505B "Proteome-wide non-cleavable crosslink identification with MS Annika 3.0 reveals the structure of the C. elegans Box C/D complex"

We thank the reviewer once more for taking the time and effort to review the revised version of our manuscript and provide feedback.

Below follows a point-by-point response to the comments made by the reviewer. The original reviewer comments are followed by our response in blue.

Reviewers' comments:

Reviewer #1

(Remarks to the Author):

The authors have addressed all my comments.

We would like to thank the reviewer for approval of our manuscript and express our gratitude for their efforts and giving such detailed feedback over the course of the past revisions!